

WESTERN SYDNEY
UNIVERSITY



Module 8

Symbolic Math Toolbox Functions

Manipulating symbolic expressions

- `numden()`
- `expand()`
- `factor()`
- `collect()`
- `simplify()`
- `poly2sym()`

numden()

- The **numden** () command is used to separate the numerator and denominator of a quotient.

Use **numden** () on the following expression:

```
>> y=2*(x+3)^2/(x^2+6*x+7)
y =
(2*(x + 3)^2)/(x^2 + 6*x + 7)
>> [numerator, denominator]=numden(y)
numerator =
2*(x + 3)^2
denominator =
x^2 + 6*x + 7
```

expand()

- **expand()** is used to expand an expression by expanding the products of factors in an expression.

- Expand the numerator of y:

```
>> expand(numerator)
ans =
2*x^2 + 12*x + 18
```

factor()

- The **factor()** command is used to factor an expression into a product of terms.

Examples:

```
>> factor(x^2+6*x+9)
```

```
ans =
```

```
[ x + 3, x + 3]
```

```
>> factor(6*x^3+31*x^2+34*x-15)
```

```
ans =
```

```
[ 2*x + 5, x + 3, 3*x - 1]
```

collect()

- The `collect(P, x)` command collects coefficients in a polynomial `P` in terms (powers) of `x`.

```
>> syms x y
>> P=3*y*x^3+2*y^2*x^2+10*y^3*x+20*x*y+5*x+4*y
P =
3*x^3*y + 2*x^2*y^2 + 10*x*y^3 + 20*x*y + 5*x + 4*y
>> collect(P,x)
ans =
3*y*x^3 + 2*y^2*x^2 + (10*y^3 + 20*y + 5)*x + 4*y
>> collect(P,y)
ans =
10*x*y^3 + 2*x^2*y^2 + (3*x^3 + 20*x + 4)*y + 5*x
```

x^0
term

y^0
term

simplify()

- The **simplify()** command uses the Maple simplification algorithm to simplify each part of an expression.
- Example:

```
>> syms a
>> b=3*a-(a+3)*(a-3)^2+a^2+5*a
b =
8*a - (a - 3)^2*(a + 3) + a^2
>> simplify(b)
ans =
- a^3 + 4*a^2 + 17*a - 27
```

Note: similar terms gets added together even without simplify

poly2sym()

- The **poly2sym()** function uses an array of coefficients to create a polynomial:

```
>> c=[1 3 2]
c =
     1     3     2
>> d=poly2sym(c)
d =
x^2 + 3*x + 2
```

- The function **sym2poly()** is the inverse of **poly2sym()**.

```
>> e=sym2poly(d)
e =
     1     3     2
```


solve()

- The **solve()** function sets an expression equal to zero, then solves the equation for its roots.

```
E1=x^2 - 9
```

```
solve(E1)
```

```
ans=
```

```
3
```

```
-3
```

Example

```
>> syms a b c x
>> solve(a*x^2 + b*x + c)
ans =
  - (b + (b^2 - 4*a*c)^(1/2)) / (2*a)
  - (b - (b^2 - 4*a*c)^(1/2)) / (2*a)
```

By default the expression is solved for **x**.
It is the same as

```
>> solve(a*x^2 + b*x + c==0)
ans =
  - (b + (b^2 - 4*a*c)^(1/2)) / (2*a)
  - (b - (b^2 - 4*a*c)^(1/2)) / (2*a)
```

Example

```
>> syms x a b c
>> solve(a*x^2+b*x+c, a)
ans =
-(c + b*x)/x^2
```

Note that this solves the expression for “a”.

It is the same as

```
>> solve(a*x^2+b*x+c==0, a)
```

Systems of equations

- You can use **`solve()`** to find the solution of a system of equations.

```
syms x y z;
```

```
eq1 = 3*x + 2*y - z == 10;
```

```
eq2 = -x + 3*y + 2*z == 5;
```

```
eq3 = x - y - z == -1;
```

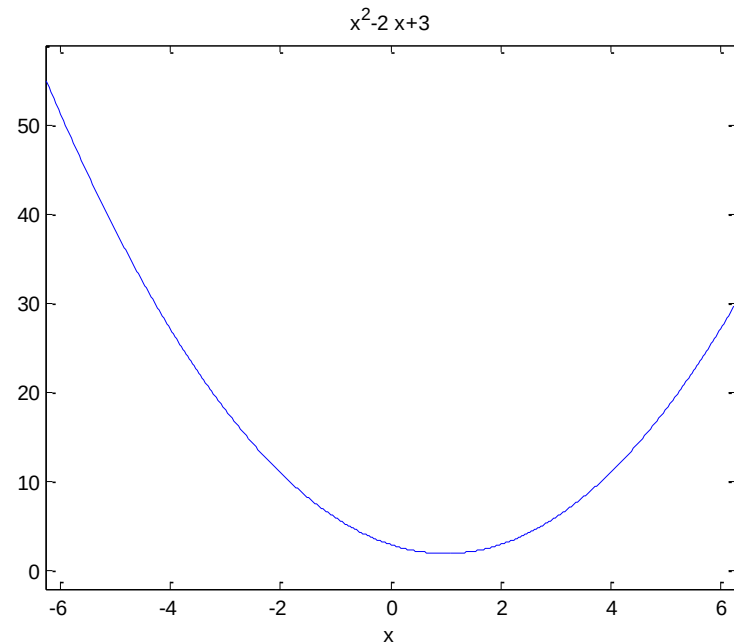
```
[x1, y1, z1]=solve(eq1, eq2, eq3)
```

Note that the solve function produces symbolic output. You can change the output to numerical values with the **`double()`** command.

Plotting symbolical functions

- Plotting symbolic functions in MATLAB is done with the `fplot()` set of commands.
- The syntax of `fplot()` when y is a function of x is:

```
y=x^2-2*x+3  
fplot(y)
```



Differentiation

- MATLAB allows you to differentiate symbolic functions with the **diff()** command.
- The command **diff(f,x,n)** gives n^{th} order derivative of f as a function of x , when f is a symbolic function of the variable x

```
>> syms x
>> f = x^3+2*x^2+3
f =
x^3 + 2*x^2 + 3
>> diff(f,x,1)
ans =
3*x^2 + 4*x
```

```
>> diff(f)
ans =
3*x^2 + 4*x
>> diff(f,x,2)
ans =
6*x + 4
```

$$\frac{df}{dx}$$

$$\frac{d^2f}{dx^2}$$

Integration

- MATLAB allows you to integrate symbolic functions with the `int()` command. Either definite integrals, with assigned numeric or symbolic bounds, or indefinite integrals (*note that when you compute indefinite integrals, the constant of integration does not appear explicitly*).

```
>> syms x
>> y = 3*x^2 + 4*x
y =
3*x^2 + 4*x
>> int(y)
ans =
x^2*(x + 2)
```

```
>> int(y,x)
ans =
x^2*(x + 2)
>> int(y,1,2)
ans =
13
```

$$\int y dx$$

$$\int_1^2 y dx$$